

Operating a Volunteer OpenSky Node in Brazil:

Field Notes on Coverage, Uptime, and Local Regulations

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KEY RESULTS

189 →
222 km

MEDIAN RANGE

Era A → Era B

+15.6%

MEAN RANGE

same 2,234 aircraft

188.5 → 217.9 km; pooled position ranges, paired windows.

99.54%

RECEIVER UPTIME

2 confirmed receiver outages
5 h 12 min, 6 July
9 h 47 min, 2 August

99.33% reception continuity

40 gaps > 120 s

19 start at 05:00 to 07:00 UTC. Receiver statistics cannot distinguish these night gaps from no observable traffic.

786 km

MAXIMUM OBSERVED RANGE

3 EVENTS > 600 km

16 to 18 June
15 August
25 August

Exact maximum: 786.3 km.
Outside event windows: 599.9 km.

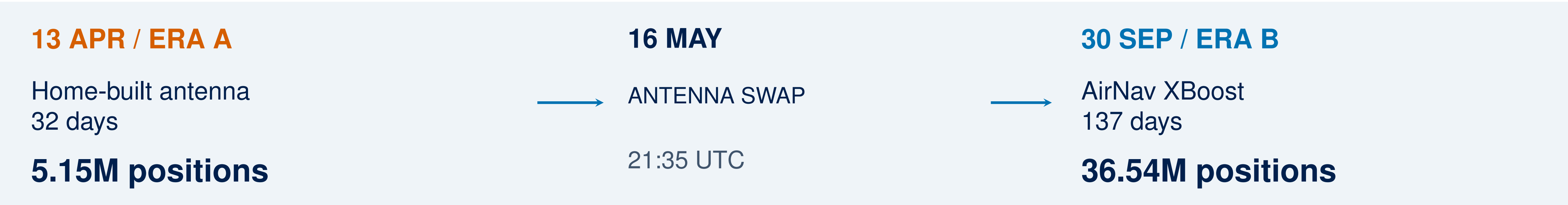
COST

US\$150 →
US\$253
+69%

Illustrative Remessa Conforme kit:
US\$150 sticker + US\$60 duty + US\$43 ICMS.
ICMS 17% gross-up. Total approximate.

One Brazilian feeder • 169 days of ADS-B observations

Coverage improved after an antenna change; interruptions had mixed operator-reported causes.

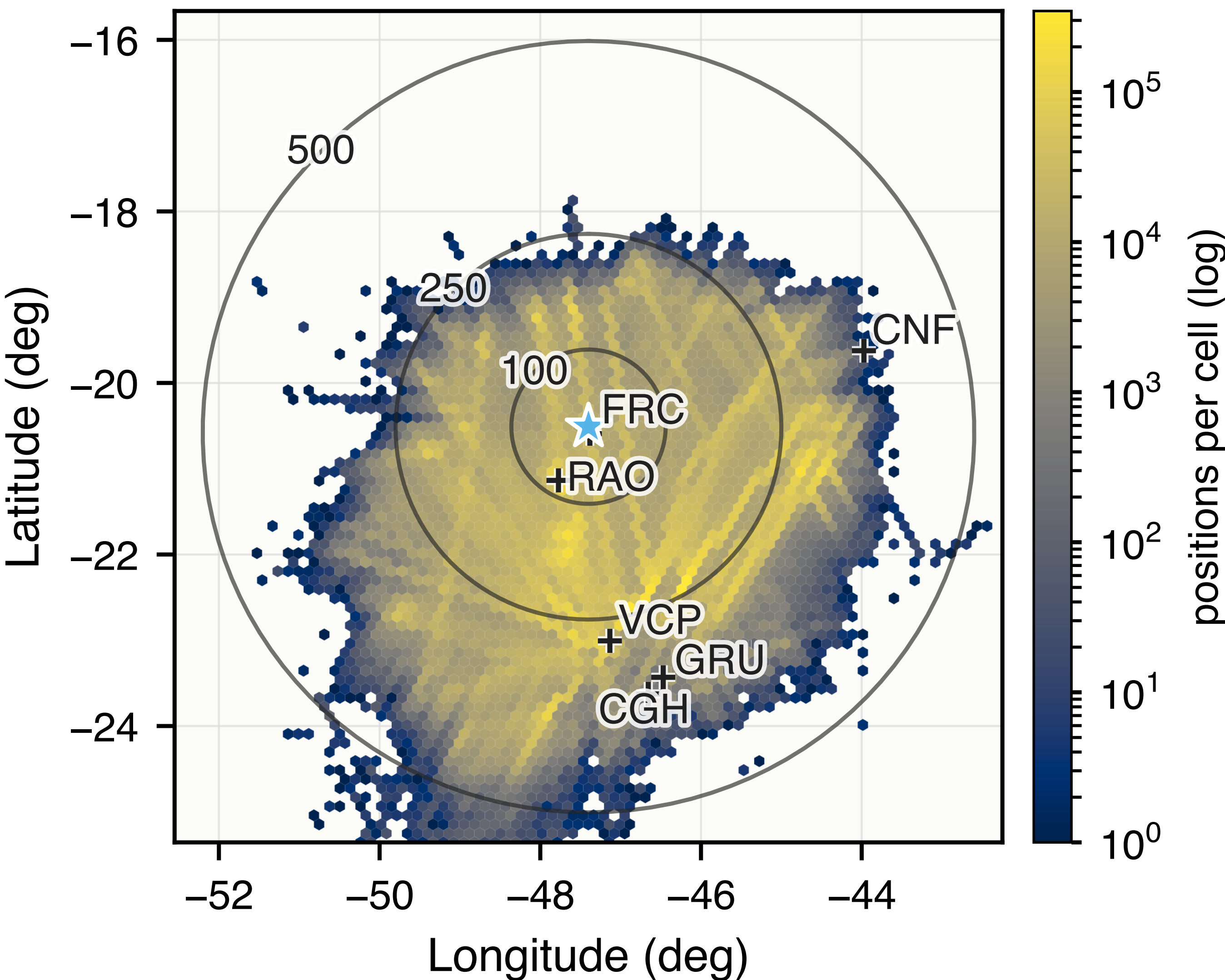


Local readsb traces • direct ADS-B only • MLAT excluded

adsb_icao / adsb_icao_nt Era A: 13 Apr to 14 May; Era B: 16 May to 30 Sep 2026.

One node: local evidence, not a network-wide OpenSky coverage claim.

COVERAGE What changed after the antenna swap?

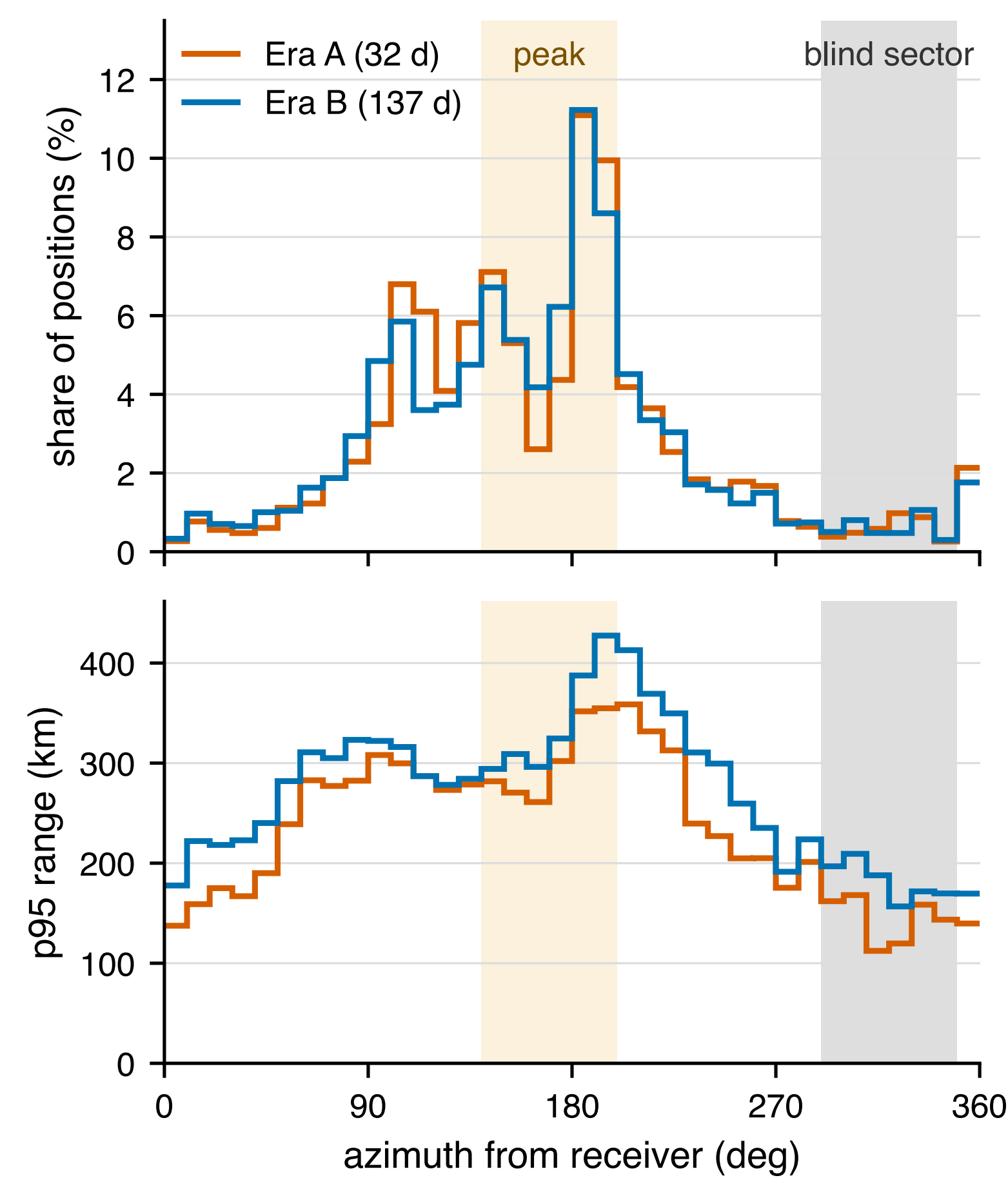


Era B density; rounded (-20.51, -47.40). Rings in km.

Era	Positions	Aircraft	Median	P95
A	5,153,167	2,890	189.3	299.7
B full	36,535,422	4,925	222.0	350.1
B paired	8,123,970	3,159	224.0	350.6

Ranges in km. Paired B: 32 days. Era A max: 510.3 km.

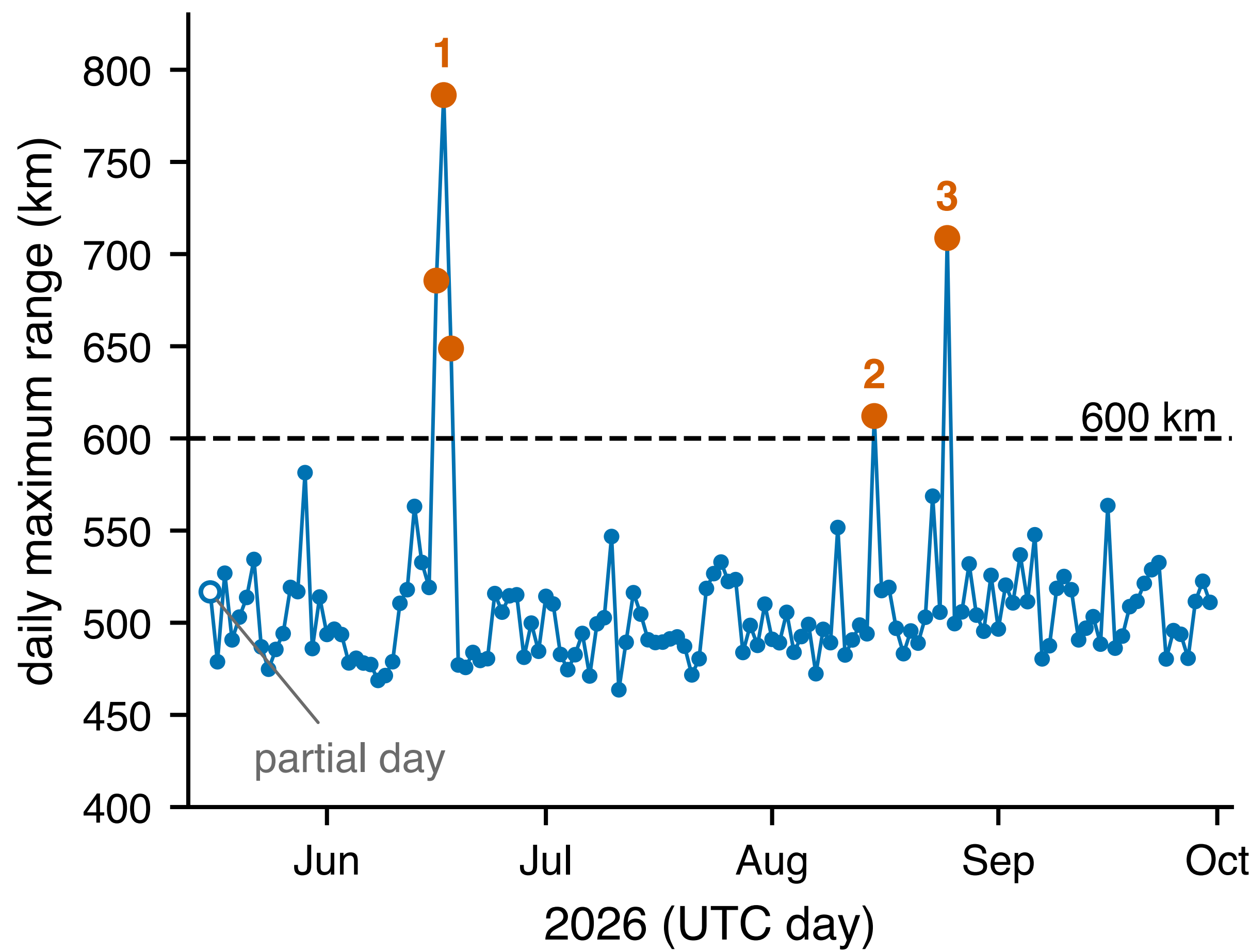
RANGE BY AZIMUTH



Blind sector 290 to 350° persisted with both antennas, pointing to local terrain/obstruction.

Peak: 140 to 200°, towards the São Paulo terminal area.

DAILY MAXIMUM RANGE



Context: Idec (2023), ANEEL (2026), ANATEL 715/2019, LGPD, DECEA, MF 1.342/2026. References via QR.

KEY TAKEAWAYS

COST

Import taxation amplifies the hardware barrier.

INFORMATION

Rules are fragmented rather than explicitly prohibitive.

POWER

Power interruptions also affected this Southeast node.

BRAZILIAN CONTEXT

INFORMATION

ANATEL equipment homologation
LGPD home-location privacy; international data transfer
DECEA no continental ADS-B Out mandate

POWER Regional context

DEC: interruption hours/consumer, 2022.

Southeast ~7 h / 4 FEC

North ~24 h / 12 FEC

Brazil 2025 DEC: 9.30 h.
FEC: interruptions/consumer.

Reported causes: power loss, k3s update conflicts and SDR overheating. No per-event attribution.

OPEN DATA REPRODUCIBILITY

Dataset v1.0.0
doi:10.5281/
zenodo.23090597

